# DBMS ASSIGNMENT 1

1. Design an E-R diagram and relational table for an IT training group database that will meet the information needs for its training program. Clearly indicate the entities, relationships and the key constraints. The description of the environment is as follows: the company has 12 instructors and can handle up to 100 trainees for each training session. The company offers 5 advanced technology courses, each of which is taught by a team of 2 or more instructors. Each instructor is assigned to a maximum of two teaching teams or may be assigned to do research. Each trainee undertakes one adanced technology course per training session.

2. Design an E-R diagram for a COMPANY database as per the requirements given below. Make appropriate assumptions to complete the specification.
   a. The company stores the information about the currently working employees. The information includes employee number, name, gender, salary, and date of birth, date of joining, address and phone number. Each employee works for a department on a particular project for a particular number of hours.
   b. The information about departments includes department number and department name. each department controls some projects currently running in the company. Also each department is managed by a particular employee who becomes the manager for that department. This employee also supervises all the other employees in that department.
   c. The project information includes project number, project name and its description.
   d. An employee can work for only one department; however, a department can have any number of employees. A department is managed by only one manager and a manager can manage only one department. A department can control any number of projects; however, one project can be handled by only one department. Any number of employees can work on any number of projects.
   Specify the key attributes of each entity type, role names and mapping cardinalities.

3. Consider a BANK database having customer, loan, account, employee, and branch as entity types. Each branch of bank allows customers to open accounts and borrow loans. A customer can open more than one account, and one account may also belong to one or more customers. Similarly, a customer can take out more than one loan and a loan may be held by more than one customer. The bank has a number of employees working in different branches of the bank. Add appropriate attributes for each entity type. Represent the key attribute, weak entity types (if any) and cardinality ratios. Make appropriate assumptions to complete the specification. Design an E-R diagram for the BANK database.

4. Consider the following relation schema for a UNIVERSITY database:
   PROFESSOR (PID, P_Name, Dept_ID)
   COURSE (Code, Dept_ID, C_Name, Syllabus)
   PROF_COURSE (PID, Code, Semester)
   Identify the primary keys and foreign keys for this schema. Now populate these relations with some tuples, and then give an example of insertion in the PROF_COURSE relation that violates the referential integrity constraints and of another that does not. Identify what you think should be the various candidate keys for this schema. Make any assumption, wherever necessary.

5. Consider the following relation schema for the SALES database:
   CUSTOMER (CustNo, Cust_Name, Address)
   ORDER (OrderNo, Order_date, Cust_no, Qty, Amount)
   PRODUCT (ProdNo, Price, Order_no)
   Specify the foreign key constraints for the SALES database. Also insert some tuples in the relations and show some examples of deletion of tuples that violate referential integrity constraints. Make any assumption, wherever necessary.

6. Implement relational algebra, calculus, and SQL queries including join operation for following schemas.

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